Which Steam tag pair should an indie studio build next? Horror × Multiplayer

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

Steam is the largest PC game store. Players vote tags onto each game; the top tags drive store search.

1 The decision

The creative director of an indie studio decides which two-tag game concept to build next. If no pair had passed the tests below, they would pick a proven but crowded pair such as Management × Simulation.

2 Why a network

Tag counts show which tags are common. They cannot show which two tags sit in separate parts of the tag network yet do well when combined; that needs each tag's neighbours.

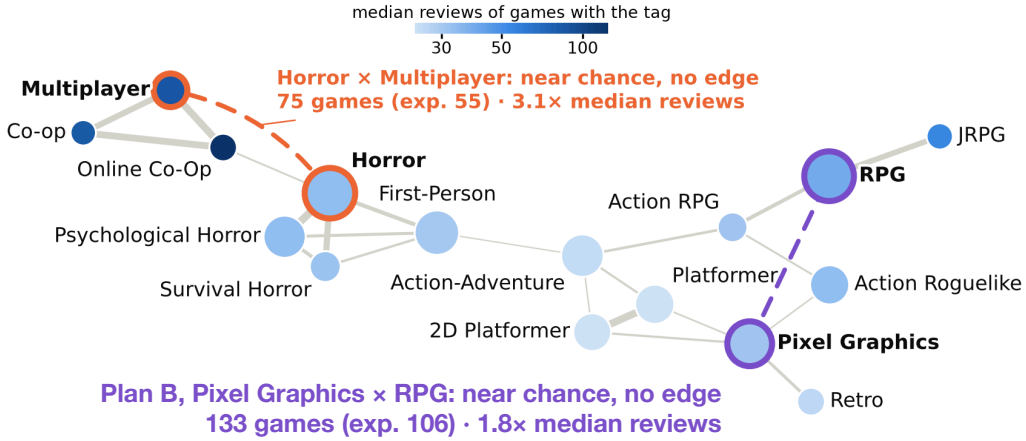
3 Data

Source	S1 FronkonGames dataset (Hugging Face). S2 live Steam store + review API.
Obtained	S1 download (Sep 2025 snapshot). S2 scraped 2026-10-14: random 1,000 (seed 7202), 1 req / 1.5 s.
Legal	S1 MIT licence. robots.txt allows /app/, /appreviews/. No personal data.
Period	Released Jan 2023 – Dec 2024.
Volume	S1: all 9,094 games with ≥10 reviews (network). S2: 996 (check).
Cross-check	S1 vs S2 top-5 tags: Jaccard 0.87; reviews ρ = 0.98.
Bias	The review floor favours games that sold; reviews count buyers, not unmet demand; late-2024 games had less time.

4 Graph construction

Node	A Steam tag (one node type). Undirected, weighted tag–tag projection of the game–tag bipartite graph.
Edge	Co-occurrence in a game's top 5 tags; suggests relatedness, not a stated relation.
Resolution	None needed: Steam tags are a fixed list, already resolved by the platform.
Weight	NPMI (0 = what chance predicts). With raw counts, 9 of the top-10 edges involve the 5 most common tags; with NPMI, 0.
Window	Top 5, as on the store page. With all 20, 200 games carry the pair, mostly as minor tags; lift falls from 3.1× to 2.5×.
Excluded	"Indie" (seed); release-model and adult tags; tags on >25% of games: Singleplayer, Adventure, Action, Casual, 2D, 3D, Atmospheric, Exploration.
Pruning	Flat cutoffs: tag on ≥1.5% of games (136); edge NPMI ≥ 0.10 and ≥0.5% shared (45); pairs scored if ≥0.8% (73).
Size	439 tags → 81 nodes, 177 edges.

5 The picture



16 of 81 tags. Orange = pick, purple = plan B, dashed = no edge (NPMI < 0.10). Size = games; width = NPMI.

Pair	Games (exp.)	Over-supply	Shared neighbours	Median reviews	Lift (÷40)	Top 10%
Horror × Multiplayer	75 (55)	1.4×	0.08	124	3.1×	24%
Pixel Graphics × RPG	133 (106)	1.3×	0.00	70	1.8×	13%
Management × Simulation	121 (32)	3.8×	0.08	127	3.2×	23%

Exp. = games chance predicts = games with A × games with B ÷ 9,094. Over-supply = games ÷ exp.

6 Findings

- **Horror × Multiplayer** is combined about as often as chance (NPMI 0.07, so no edge), and the tags share 8% of their neighbours. Yet its median game gets 3.1× the cohort's 40 reviews ($p < 0.001$).
- 9 of the 10 highest-lift pairs are crowded, e.g. Management × Simulation (3.8× chance).
- #1 at all 8 NPMI cutoffs. Lift > 1 in 100/100 bootstraps (90% CI 1.9–7.8×) and on S2 (1.6×). #1 in 36/100; in 42 it dips under the 73-game floor.

7 Recommended action

- Week 0: pitch Horror × Multiplayer at greenlight; plan B is Pixel Graphics × RPG.
- Weeks 1–6: prototype. Week 7: Coming Soon page with Horror, Multiplayer as first tags.
- Before full production: re-run on 2025 releases; plan B if over 1.5× crowded or out of the top 3.

8 What would change my mind

At 1.37× chance the pair is near the 1.5× crowding limit (crossed in 17/100 resamples). If 2025 releases cross it, I switch to plan B.